



General Orbit

M-100 Space Magnetometer

INTERFACE CONTROL DOCUMENT

REVISION HISTORY

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LIST OF FIGURES

UNITS

Unless otherwise specified in this document, all values follow the conventions below.

Quantity	Units	Symbol
Length	millimeter	mm
Mass	gram	g
Voltage	volts DC	VDC
Current	milliampere	mA
Power	milliwatt	mW
Temperature	degrees Celsius	°C
Magnetic field	nanotesla	nT
Frequency / sample rate	hertz	Hz
Data rate	bits per second	bps

Time	seconds, milliseconds	s, ms
Angle	degrees	°

LIST OF ACRONYMS & ABBREVIATIONS

ADCS – Attitude Determination and Control System
APID – Application Process Identifier
CAD – Computer-Aided Design
CCSDS – Consultative Committee for Space Data Systems
CRC – Cyclic Redundancy Check
DREQ – Data Request
EMC – Electromagnetic Compatibility
EMI – Electromagnetic Interference
ESD – Electrostatic Discharge
GEO – Geostationary Earth Orbit
GSFC – Goddard Space Flight Center
HEO – High Earth Orbit
ICD – Interface Control Document
LBNL – Lawrence Berkeley National Laboratory
LDO – Low-Dropout (regulator)
LEO – Low Earth Orbit
LET – Linear Energy Transfer
MCU – Microcontroller Unit
NED – North-East-Down
OBC – On-Board Computer
PCB – Printed Circuit Board
RX – Receive
SEE – Single-Event Effects
TID – Total Ionizing Dose
TX – Transmit
UART – Universal Asynchronous Receiver/Transmitter
VDC – Volts Direct Current

1. INTRODUCTION

1.1. PURPOSE AND SCOPE

This document defines the interface requirements for the General Orbit M-100 Space Magnetometer. It provides the information necessary for satellite integrators to correctly interface with the M-100, covering electrical connections, mechanical mounting, and software communication. This document does not cover environmental qualification, radiation test data, or declared materials lists, these are available upon request. General Orbit cannot guarantee nominal operation of the M-100 if the specifications provided in this document are not adhered to.

1.2. PRODUCT OVERVIEW

The M-100 is a three-axis space magnetometer built on magneto-inductive sensor technology. It delivers 1.5 nanotesla resolution across a $\pm 100,000$ nT measurement range at 45 milliwatts and 60 grams (without enclosure), operating on a +3.3-5 VDC supply. The M-100 is qualified for radiation environments encountered in LEO, high Earth orbit, and cislunar space, and is suitable for platforms ranging from CubeSats to large GEO satellites. It is assumed that the reader is already familiar with the M-100 product description prior to using this document.

This version of the ICD applies to the products and versions described in Table 1.

Table 1: Interface Control Document Applicability

Product	Version	Notes

1.3. REFERENCE DOCUMENTS

The documents outlined in Table 2 are referenced in this document.

Table 2: Reference Documents

Document ID	Name	Version
CCSDS RECOMMENDED STANDARD FOR SPACE PACKET PROTOCOL	CCSDS 133.0-B-2 JUNE 2020	

2. ELECTRICAL INTERFACE

2.1. COMMUNICATION INTERFACE

2.1.1. UART

The M-100 supports serial communication using the UART protocol as a single-ended physical layer. UART is recommended for benchtop testing and simple flight harnesses where one M-100 is connected to one host over short cable runs. The UART variant is supplied with a 9-pin Micro-D connector by default; alternate connector options are available upon request and shall be specified at order placement.

The M-100 includes level-shifting circuitry that shifts the logic-level high voltage of the UART signals to the supply voltage provided to the board, allowing the UART interface to interoperate with hosts operating at voltage logic levels within the supply voltage requirements without external translation. UART characteristics are listed in Table 3.

Table 3: UART Characteristics

Parameter	Value
Baud rate	115200
Data Bits	8
Parity	None
Stop bits	1
Flow control	None

2.1.2. RS-422

The M-100 supports RS-422 as a differential, point-to-point physical layer for longer cable runs and improved noise immunity. RS-422 characteristics are listed in Table 4.

Table 4: RS-422 Characteristics

Parameter	Value
Baud rate	115200
Data Bits	8
Parity	None
Stop bits	1
Flow control	None
Termination	Not required for SmallSats 150 Ω at receiver (longer connections)

2.2. POWER INTERFACE

2.2.1. SUPPLY VOLTAGE REQUIREMENTS

The M-100 accepts a DC supply voltage between 3.1 V and 5.5 V. The supply voltage range is bounded by the onboard LDO regulator. Supply voltage requirements are listed in Table 5, and power consumption in Table 6.

Table 5: Supply Voltage Requirements

Parameter	Min	Max	Unit
Supply Voltage (Vs)	3.1	5.5	VDC
Average Power	0.01	0.16	W

Table 6: Power Consumption

Voltage (V)	Average Power, UART (mW)		Average Power, RS422, Rt=0ohm (mW)		Average Power, RS422, Rt=150ohm (mW)	
	Idle	150 Hz	Idle	150 Hz	Idle	150 Hz
3.3	9	18	14.5	36.5	14.5	87.2
5	13.75	27.5	22	55	22	132

2.2.2. POWER PROTECTION & RECOMMENDATIONS

General Orbit recommends that the host platform monitor the supply voltage delivered to the M-100 to verify it stays within the limits defined in Table X (Section 2.2.1). The host should also enforce a current limit on the supply line to bound fault current in the event of a latch-up or short. The M-100 includes reverse voltage protection circuitry up to max supply voltages listed in Table X.

2.2.3. ENABLE LINE

Power-on of the M-100 is controlled by an external active-high enable line driven by the host. When the enable line is asserted, the M-100 powers up and begins normal operation. When the line is held low or left floating, power consumption of the sensor is minimized and the M-100's MCU is fully isolated from its inputs, preventing parasitic operation from voltage present on the TX, RX, or DREQ lines. Electrical characteristics for the enable line are given in the connector pinout (Section 2.3).

2.2.4. POWER AND SIGNAL GROUND

Power and signal ground are common on the M-100; a single ground reference is shared across all circuits.

2.3. CONNECTOR PINOUT

2.3.1. UART

The M-100 UART variant uses a 9 pin Micro-D connector for power, ground, and serial communication. The pinout and electrical characteristics are listed in Table 7.

Table 7: Pinout and Electrical Characteristics, UART

Pin #	Symbol	Parameter	Type	Min	Max	Units
	Vs	Supply Voltage	Power	3.3	5.0	V
	GND	System ground	Power	-	-	V
	TX	UART Transmit line	Output	0	Vs	V
	RX	UART Receive line	Input	0	Vs	V
	EN	Enable line	Input	0	Vs	V
	DREQ	Data Request line	Input	0	Vs	V

2.3.2. RS422

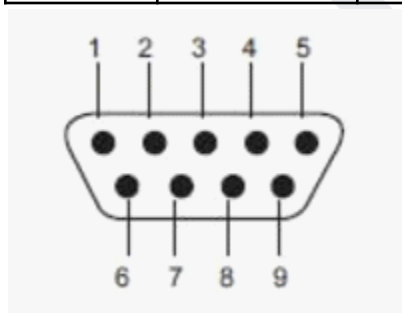
The RS-422 variant of the M-100 mates to the host harness via a 9-pin female Micro-D connector. Connector part details are listed in Table X, and the pinout and electrical characteristics are listed in Table Y.

Table 8: Connector Part Details

Part	Description	Part Number
Header	9-pin female Micro-D	
Mating housing	9-pin male Micro-D	

Table 9: Pinout and Electrical Characteristics, RS422

Pin #	Symbol	Parameter	Type	Min	Max	Units
	Vs	Supply Voltage	Power	3.3	5.0	V
	GND	System ground	Power	-	-	V
	Y	RS422 non inverting output	Output	0	Vs	V
	Z	RS422 inverting output	Output	0	Vs	V
	B	RS422 inverting input	Input	-25	25	V
	A	RS422 non inverting input	Input	-25	25	V
	EN	Enable line	Input	0	Vs	V
	DREQ	Data Request line	Input	0	Vs	V



2.4. ESD PRECAUTIONS

The M-100's external interfaces are protected by integrated ESD suppression diodes rated to ± 25 kV contact discharge and ± 30 kV air-gap discharge. This protection covers all inputs and outputs exposed at the connector. Standard ESD handling practices remain required despite this protection to safeguard internal components not directly exposed at the harness interface.

2.5. GROUNDING

A single ground connection is established at the connector, where the host harness ties the M-100's signal and power ground to the spacecraft ground reference. This topology eliminates ground loops between the M-100 and the host platform and prevents circulating return currents through the enclosure.

The host platform shall provide a single, low-impedance ground return to the M-100 through the connector ground pin(s) defined in Section 2.3.

The host platform may optionally bond the M-100 enclosure to spacecraft chassis ground if required by the spacecraft grounding architecture. Bonding shall be made through the mounting hardware to a clean, conductive chassis surface. Because the enclosure is internally isolated from the PCB ground, bonding the enclosure to chassis ground does not create a parallel return path.

If full enclosure isolation from spacecraft chassis ground is required for a specific mission, the M-100 may be mounted using non-conductive isolation hardware (insulating washers and bushings under the mounting screws). This is a host-side integration choice and does not require any modification to the M-100.

3. MECHANICAL INTERFACE

3.1. PHYSICAL PROPERTIES

The M-100 is housed in an aluminum 6061-T6 enclosure, dimensions are ___ x ___ x ___.

3.1.1. MASS

Table 10: Mass Properties

Configuration	Mass	Tolerance	Method
M-100 (without enclosure)	5 g	±5%	Measured
M-100 (with enclosure)	170 g	±5%	Measured

3.1.2. THERMAL

The M-100 operating temperature range is listed in Table 11. Mounting may be conductive or adiabatic depending on host platform thermal architecture.

Table 11: Thermal Characteristics

Parameter	Value	Notes
Operating temperature range	-40° C to +85° C	Nominal performance specifications met across this range

3.2. MOUNTING

The M-100 mounts to the host platform via four (4) threaded mounting holes located at the corners of the 100 mm × 100 mm enclosure footprint. Non-ferrous fasteners (e.g., titanium or non-magnetic stainless) shall be used to avoid introducing magnetic disturbances near the sensor.

Sufficient clearance shall be provided around the 9-pin Micro-D connector to allow reliable mating and demating of the harness. The M-100 may be mounted internally or externally on the host platform; internal mounting is preferred for thermal stability, while external mounting is supported with appropriate thermal isolation per Section 3.1.2.

The mounting hole pattern, exact hole locations, and full mechanical drawings are supplied to customers in the M-100 mechanical CAD package.

[Need visual mounting CAD figure]?

3.3. COORDINATE SYSTEM

The M-100 measurement coordinate system is NED, with the North axis pointing from the center of the enclosure to the connector, the North-East plane parallel with the lid of the enclosure, and the down axis pointing into the enclosure when viewed from the top. The origin and axis orientations are defined relative to the mounting face and connector location, as shown in Figure X .

[need CAD figure]

Magnetic field measurements output by the M-100 are reported in this frame. Any rotation between the M-100 measurement frame and the host spacecraft body frame shall be characterized and applied by the host ADCS/OBC.

3.4. MAGNETIC CLEANLINESS REQUIREMENTS

To maximize M-100 performance, the host platform should observe the following:

- Avoid mounting the M-100 in direct contact with or immediately adjacent to high-current conductors, switching power supplies, reaction wheels, magnetorquers, or permanent magnets. [julio, what is minimum recommended standoff distance.]
- The M-100 measurements should be disregarded during magnetorquer operation when uncontaminated data is required. For continuous measurement during magnetorquer activity, the host shall telemeter the commanded dipole to the M-100 for cancellation.
- Fasteners, brackets, and mounting hardware in the immediate vicinity of the M-100 shall be non-ferrous (see Section 3.2).
- Time-varying currents on harnesses routed near the M-100 should be minimized.

For missions with unusually severe magnetic environments, contact General Orbit for mission-specific characterization.

3.5. EXTERIOR SURFACE

The M-100 enclosure is machined from aluminum 6061-T6 and finished with a chromate conversion coating (MIL-DTL-5541 Class 3) to provide corrosion resistance, electrical conductivity for grounding and EMI performance, and a low-outgassing surface compatible with vacuum environments.

The exterior surface emissivity and absorptivity are listed in Table X. These values shall be used by the host platform for thermal analysis. Custom surface treatments (e.g., black anodize, alodine alternatives, or thermal coatings) are available upon request and shall be specified at order placement.

Table 12: Exterior Surface Properties

Parameter	Value	Other
Material	Aluminum 6061-T6	
Surface treatment	Chromate conversion coating	MIL-DTL-5541 Class 3

Solar absorptivity (α)		
Infrared emissivity (ϵ)		

The exterior of the M-100 shall not be painted, labeled, or otherwise modified by the customer without prior coordination with General Orbit, as surface modifications can affect thermal performance.

A serial number label is affixed to one face of the enclosure. The label location is shown in the mechanical CAD package and shall be considered when planning host-side mounting and accessibility.

4. SOFTWARE INTERFACE

This section defines the software interface to the M-100, including the supported communication protocols, operating modes, telemetry packet structure, and command set. All commands and responses use a binary message format with a CRC-16 checksum for command and response integrity.

4.1. SUPPORTED PROTOCOLS

The M-100 communicates with the host over a serial connection using either UART or RS-422 (selected at order placement, see Section 2.1). The application-layer protocol is identical across both interfaces and consists of a binary command/response structure with CRC-16 error checking. The M-100 does not initiate communication except in continuous mode, where data packets are streamed at the configured sample rate without host prompting.

4.1.1 MSBP

The M-100 uses a proprietary binary packet protocol named MSBP. MSBP packets consist of the following: optional two byte sync word, one byte message ID, payload, and two byte CRC16 checksum. Packets transmitted by the host to the M-100 are named Command messages and the sensor packets sent back in response are named Response messages. Commands come in get/set pairs when possible (example exception: get data does not have a corresponding set data command).

Certain bits in the Message IDs are used for different purposes. Bit 7 (0x40) is set if the command is a get command. Bit 8 (0x80) is set if the command is a response command. Get commands are used to read out information from the M-100. All set commands have corresponding get commands at base message ID | 0x40. All get commands have corresponding responses at base message ID | 0x80. All get commands have zero length. Get commands with corresponding set commands are omitted here for brevity.

Commands shall be sent one at a time; the host shall wait for the response to a command before issuing the next. The M-100 does not buffer commands beyond the currently executing one. In continuous mode, telemetry packets may arrive between a command being sent and its response being received; the host shall be prepared to parse interleaved telemetry and command responses.

All host-initiated commands receive a response. All commands and responses have fixed length corresponding to their message ID except for the multisample data response. All commands have corresponding responses. If an error occurs while processing a command message, the M-100 will respond with an error response. Command messages with invalid message IDs, valid message IDs but incorrect length, or invalid checksums are responded to by the M-100 with an error response. Command messages not received in full in 250ms will be responded to with an error response as well. Response messages from the M-100 with invalid checksums should be discarded.

A complete list of supported commands, their parameters, and response formats is provided in Appendix A. The following sections define the message format, checksum, and high-level command categories.

4.1.2 CCSDS

The M-100 can be configured to support the CCSDS Space Packet Protocol as a higher-layer protocol over MSBP. CCSDS support is enabled via the CCSDS configuration set command (Section 4.4.3).

When CCSDS mode is enabled, the M-100 transmits responses as CCSDS Space Telemetry Packets and accepts only command packets from the host. Any telemetry packets received by the M-100 are rejected with an error response. The M-100 transmits only unsegmented telemetry packets and does not accept segmented command packets; segmented packets received by the M-100 are rejected with an error response.

CCSDS packets transmitted by the M-100 consist of the CCSDS primary header followed directly by the appropriate MSBP response. The M-100 does not implement a CCSDS secondary header and does not include onboard timestamps in the packet structure. Hosts requiring time-tagged measurements shall apply timestamps at the receiving end based on packet arrival time or correlate against a host-maintained time reference.

The M-100's default Application Process Identifier (APID) is 0x7D0. The APID is reconfigurable via the CCSDS configuration set command and is persisted in non-volatile memory across power cycles.

4.1.3 PACKET FORMAT

MSBP Packets have the following format (except for the Multisample data response):

- Optional 2 byte syncword
- 1 byte message ID
- Fixed length payload (length zero for get commands)
- 2 byte checksum

CCSDS Packets have the following format (including the Multisample data response):

- 6 byte CCSDS Header
 - 3 bit packet version number (value 0b000)
 - 1 bit packet type (always value 0, telemetry)
 - 1 bit secondary header flag (always value 0, no secondary header)
 - 11 bit APID (User configurable, default 0x7D0)
 - 2 bit sequence flag (always 0b11, unsequenced data)
 - 14 bit sequence count (always 0)
 - 2 byte packet data length (value: MSBP payload length + 2)
- 1 Byte MSBP Message ID
- Fixed length MSBP payload
- 2 byte checksum

4.1.4 MULTISAMPLE DATA RESPONSE

The Multisample data response has the following format. It differs from the other MSBP responses because it can have variable length. The CCSDS packet header includes payload length however so the Multisample data response does not differ from the format given in section 4.1.3.

- Optional 2 byte syncword
- 1 byte message ID
- 2 byte payload length
- variable length payload
- 2 byte checksum

4.1.5 CHECKSUM

All M-100 MSBP and CCSDS packets contain a checksum appended to the end of the message. The checksum is computed over all bytes from the start of the message through the end of the payload, excluding the checksum bytes themselves.

The checksum is CRC16 CCITT (0xFFFF)

4.2. OPERATING MODES

The M-100 supports two operating modes: continuous and single measurement. The default mode at power-on is single mode; the default mode can be changed to continuous mode via the command interface.

4.2.1. CONTINUOUS MODE

In continuous mode, the M-100 takes magnetic field measurements at the configured sample rate and sends data response messages to the host without host prompting. The default sample rate is 150 Hz, configurable down to lower rates via the command interface for power-constrained applications.

Commands sent to the M-100 while in continuous mode will pause measurement streaming until the command is processed and responded to. Command messages can take as long as 250ms to process, so they should be done sparingly to avoid excessive data loss. Hosts requiring deterministic sample timing should use single mode with hardware-triggered measurements via the DREQ line.

4.2.2. SINGLE MODE

In single mode, the M-100 remains in a low-power state until it receives a measurement request. Requests can be issued either via a get data command over the serial interface or via a rising edge on the DREQ line (Section 2.3). On receipt of a request, the M-100 takes a single measurement, returns the result in a response telemetry packet, and returns to low-power state.

Single mode is recommended for power-constrained missions, applications requiring synchronized sampling with other spacecraft sensors, or operation at sample rates not supported through the sample rate configuration command. The max sample rate that can be achieved using single measurement mode is lower than the continuous mode maximum due to various sources of overhead intrinsic to manual operation of the sensor via serial commands or the DREQ pin and trade offs made to reduce power consumption between measurement requests.

4.3. CONFIGURATION

Configuration parameters — including sample rate, packet rate, enabled sensors, and operating mode — can be modified via the command interface (Section 4.4.1). A subset of configuration parameters is persisted in non-volatile memory and survives power cycles; the remaining parameters revert to factory defaults at power-on. Per-parameter persistence behavior is documented in the configuration command reference in Section 4.4.2.

4.3.1. SENSOR SETTINGS

The following sensor settings are available for modification by the host through the set M config command: mode, enable flags for individual mags 1-4, temperature sensor, and data request line. The set M config command can also be used to clear hardware malfunction flags for sensors the M-100 has deemed nonoperational (timeout based). Sensors that are not enabled will not be used. Sensors that are marked as nonoperational through the malfunction flags will not be used. The host can clear the malfunction flags but there is no guarantee nonoperation sensors can be recovered through this functionality.

4.3.2. PACKET PERIOD

The M-100 supports sending data responses at periodic intervals configured by the user. The number of samples per packet equals the configured sample period in seconds multiplied by the configured sample rate in Hz. The M-100 sensor supports up to a two second packet period at the max 150 Hz sample rate

4.3.3. SAMPLE RATE

The continuous mode sample rate of the M-100 can be configured to the following fixed amounts: 150 Hz, 75 Hz, 37 Hz, 18 Hz, 9 Hz, 4.5 Hz, 2.3 Hz, 1.2 Hz, 0.6 Hz, 0.3 Hz, 0.015 Hz, 0.075 Hz. All sample rates are approximate due to the nature of the M-100's magnetometer technology.

4.3.4. CCSDS

The CCSDS set command can be used to set the APID of the M-100 and to enable CCSDS support. The APID configured through use of this command is persistent between power cycles.

4.4. REFERENCE

4.4.1. COMMANDS

Command messages are packets sent to the M-100 from the host.

Command	Set M Configuration	
Description	Change mode, Enable disable hardware features, clear hardware malfunction flags	
Comment	Can take up to 250ms to process.	
Message structure	Message ID	Length (bytes)

	0x01	5		
Byte offset	Number format	Name	Unit	Description
0	U1	Mode	Single, Continuous	Mode of the M-100
1	F2	Hardware Enable flags	-	Flags used to enable/disable sensors and hardware features (like data request line)
3	F2	Hardware malfunction clear flags	-	Flags used to clear hardware malfunction flags

Hardware enable/Malfunction flags

Bit 0	B1 enable
Bit 1-3	RESERVED (leave 0)
Bit 4	DREQ enable
Bit 5-15	RESERVED (leave 0)

Command	Set Packet Period			
Description	Set packet period of the M-100 data responses in continuous mode.			
Comment	Set to 0 to disable manual packet period functionality. Max packet period: 2s			
Message structure	Message ID	Length (bytes)		
	0x02	2		
Byte offset	Number format	Name	Unit	Description
0	U2	Packet Period	2048 LSB/s	Packet period

Command	Set Sample Rate			
Description	Set sample rate of M-100 in continuous mode			

Comment	Can take up to 250ms to process.			
Message structure	Message ID	Length (bytes)		
	0x03	1		
Byte offset	Number format	Name	Unit	Description
0	U1	Sample rate	-	Value corresponding to desired sample rate. VALUES SET THAT ARE NOT IN THE SAMPLE RATE TABLE WILL BE IGNORED.

Sample Rate Value (U1)	Sample Rate (Hz)
0x00	150
0x01	75
0x02	37
0x03	18
0x04	9
0x05	4.5
0x06	2.3
0x07	1.2
0x08	0.6
0x09	0.3
0x0A	0.015
0x0B	0.075

Command	Get Data	
Description	Manual data measurement initiation command	
Comment	-	
Message structure	Message ID	Length (bytes)
	0x45	0

Command	Get status command	
Description	Request status response	
Comment	-	
Message structure	Message ID	Length (bytes)
	0x46	0

Command	Get Revision ID command	
Description	Request firmware revision response	
Comment	-	
Message structure	Message ID	Length (bytes)
	0x47	0

4.4.2. RESPONSES

All commands and responses use the following message structure:

Response	M Configuration			
Description	Response containing mode, Enabled hardware features, hardware malfunction flags information			
Comment	Can take up to 250ms to process.			
Message structure	Message ID	Length (bytes)		
	0x81	5		
Byte offset	Number format	Name	Unit	Description
0	U1	Mode	Single,Continuous	Mode of the M-100
1	F2	Enabled Hardware flags	-	Flags that show which hardware features are enabled
3	F2	Hardware malfunction flags	-	Flags that show which hardware features have been marked as nonoperational by the M-100 firmware

Response	Packet Period			
Description	Response containing packet period information			
Comment	Set to 0 to disable manual packet period functionality. Max packet period: 2s			
Message structure	Message ID	Length (bytes)		
	0x82	2		
Byte offset	Number format	Name	Unit	Description
0	U2	Packet Period	2048 LSB/s	Packet period

Command	Sample Rate			
Description	Response containing sample rate information			
Comment	Can take up to 250ms to process.			
Message structure	Message ID	Length (bytes)		
	0x83	1		
Byte offset	Number format	Name	Unit	Description
0	U1	Sample rate	-	Value corresponding to desired sample rate

Response	Data			
Description	Data measurement response			
Comment	-			
Message structure	Message ID	Length (bytes)		
	0x85	14		
Byte offset	Number format	Name	Unit	Description
0	24Q8	Bx	nT	B x component
4	24Q8	By	nT	B y component
8	24Q8	B1z	nT	B z component

12	i2	Temperature	Deg C	Temperature reading

Response	Status			
Description	Response containing M-100 configuration information and the error code corresponding to the latest error that occurred			
Comment	Used as error response when an error occurs while processing commands or generating other responses.			
Message structure	Message ID	Length (bytes)		
	0x86	6		
Byte offset	Number format	Name	Unit	Description
0	U1	Mode	Single,Continuous	Mode of the M-100
1	F2	Enabled Hardware flags	-	Flags that show which hardware features are enabled
3	F2	Hardware malfunction flags	-	Flags that show which hardware features have been marked as nonoperational by the M-100 firmware
5	U1	Latest Error Code	-	Code corresponding to the latest error that occurred

Response	Multisample data
Description	Response used when in continuous mode and the packet period is not 0. Replaces the data response
Comment	Variable length response,

Message structure	Message ID	Length (bytes)		
	0x87	2 + number of samples*50		
Byte offset	Number format	Name	Unit	Description
0	U2	Payload length	Bytes	Length of the rest of the data in this response
1	-	Payload	-	N data samples formatted as in the data response stored sequentially (sample_1, sample_2,..., sample_n).

5. ENVIRONMENTS

This section defines the environmental conditions under which the M-100 has been qualified for operation, as well as the environments the unit is suitable for based on its qualification profile and flight heritage. Environmental qualification beyond what is described in this section is available as a custom option and shall be discussed with General Orbit at order placement.

The M-100 is suitable for operation in low Earth orbit, high Earth orbit, geostationary orbit, cislunar space, lunar orbit, lunar surface (with host-provided thermal accommodation for lunar night), and Mars orbit. Suitability for other environments — including Mars surface, deep space beyond 1 AU, and the Jovian radiation environment — is available upon request and may require mission-specific qualification.

5.1.1. QUALIFIED ENVIRONMENTS

The M-100 critical components have been qualified to the environmental conditions listed in Table Y. Qualification testing has been performed at NASA Goddard Space Flight Center, Lawrence Berkeley National Laboratory, and the University of Michigan. Test reports are available to customers under NDA.

Table 16: Environmental Qualification Levels

Parameter	Qualification Level	Notes
Total Ionizing Dose (TID)	>150 kRad	Tested at NASA GSFC and University of Michigan
Single-Event Effects (SEE)	LET 76.7 MeV·cm ² /mg, no destructive events	Destructive testing at LBNL 88-Inch Cyclotron
Operating temperature	-40 °C to +85 °C	Nominal performance specifications met across this range
Survival temperature		-
Thermal cycling	Per GSFC-STD-7000	-40 °C to +85 °C
Random vibration	Per GSFC-STD-7000 protoflight	-
Sine vibration	Per GSFC-STD-7000	Required if fundamental frequency <100 Hz per GSFC-STD-7000
Shock	Not expected to be a concern for the M-100	Shock testing available per mission-specific request
EMI/EMC	MIL-STD-461G in progress	Targeted Completion, July 2026
Vacuum	TBD	TBD

The M-100's radiation qualification profile supports operation in LEO, HEO, GEO, cislunar space, lunar orbit, and on the lunar surface. The M-100 has flight heritage on the Artemis Lunar Gateway HERMES platform (NEMISIS instrument), the Geospace Dynamics Constellation, and is manifested for the NASA Resolute sounding rocket mission (Andøya, Norway, 2027).

For lunar surface deployment, the lunar-night extreme (~-170 °C) exceeds the M-100's current operational and survival envelope. Host-provided thermal accommodation — including internal mounting, thermal isolation, and survival heater provisioning (2.5 W worst-case lunar-night exterior mount) — is required for missions extending through lunar night. Lunar-surface-specific thermal qualification is in progress at the University of Michigan and is on track for completion in June 2026.

Mission profiles that exceed the qualified environments listed in Table Y require additional mission-specific qualification. General Orbit can perform extended qualification testing as a custom engineering effort; contact General Orbit for scoping and timeline.